

4 The time of death of a mammal can be determined by a number of different methods.

(a) Insects can be used to determine the time of death of a mammal.

(i) Which is the term used to describe the use of insects in determining the time and place of death of a mammal?

(1)

- ☐ **A** chromatography
- ☐ **B** dendrochronology
- ☐ **C** forensic entomology
- ☐ **D** heterozygosity

(ii) Which of the following factors provides information about the time of death of a mammal?

1. species of insect
2. size of larvae, e.g. maggots
3. stages in development of insects

(1)

- ☐ **A** 1 only
- ☐ **B** 1 and 3 only
- ☐ **C** 2 and 3 only
- ☐ **D** 1, 2 and 3

(b) The body temperature of a human corpse can be used to determine the time of death.

(i) Which is the most accurate method of measuring the body temperature of a human corpse to determine time of death?

(1)

- ☐ **A** placing a temperature-sensitive strip on the forehead
- ☐ **B** pushing a temperature probe into the liver
- ☐ **C** putting a thermometer under the tongue
- ☐ **D** putting a temperature probe under the armpit

(ii) Time of death can be estimated by determining the fall in body temperature.

Which row of the table shows how more clothing and higher ambient temperature would affect the estimated time of death?

(1)

		Effect on estimated time since death of:	
		more clothing	higher ambient temperature
☐	A	increase	increase
☐	B	increase	decrease
☐	C	decrease	increase
☐	D	decrease	decrease

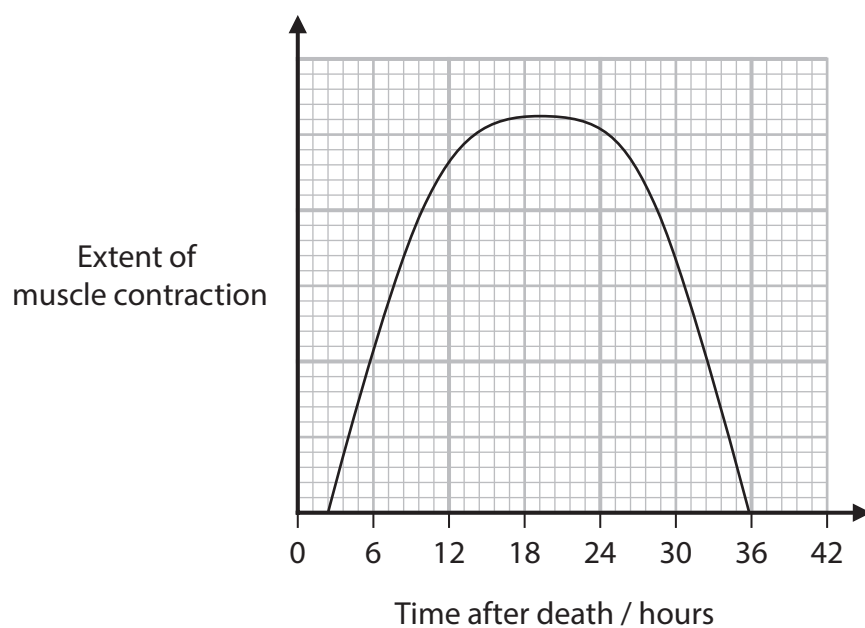
(i) The graph shows the changes in three characteristics of muscle with time after the death of a mammal:

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- Changes in the characteristics of muscle
- | Time after death / hours | ATP (solid line) | pH (dotted line) | Lactic acid (dashed line) |
|--------------------------|------------------|------------------|---------------------------|
| 0 | High | High | Low |
| 1 | High | Medium-High | Low-Medium |
| 2 | High | Medium | Medium-Low |
| 3 | Medium-High | Medium-Low | Medium |
| 4 | Low | Low-Medium | Medium-High |
| 5 | Low-Medium | Low | High |
| 6 | Low | Low | Very High |
| 7 | Low | Low | Very High |
- Time after death / hours
- lactic acid
- pH
- ATP

Use your own knowledge to support your answer.

(3)

- (ii) The graph shows the changes in the extent of muscle contraction with time after death.



Explain why using the extent of muscle contraction by itself may give an incorrect time of death.

Use the information in the graph to support your answer.

(3)

(Total for Question 4 = 10 marks)